

Hoang Vu

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unias | Portfolio



Education

Saigon University

Expected 2027

B.S. in Software Engineering, Faculty of Information Technology

Ho Chi Minh City, Vietnam

- **Current Status:** 3rd-Year Student (Seeking a full-time Internship with opportunities for long-term employment).
- **Coursework:** Data Structures & Algorithms, Database Systems, Software Architecture.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS

Frontend: React.js, Next.js, Tailwind CSS

Backend: FastAPI, Node.js, NestJS

Databases: PostgreSQL, MySQL

Tools & Architecture: Git/GitHub, Docker, Postman, REST APIs, JWT, CI/CD

Selected Projects

Warehouse Management System (WMS) | Live Demo | GitHub

Jan 2026 – June 2026

Tech: Laravel, PHP, PostgreSQL, Tailwind

- Architected a comprehensive warehouse management system supporting centralized inventory tracking across a dynamic location hierarchy (zones, racks, shelves, and bins).
- Designed a PostgreSQL schema to manage products, batches, suppliers, and internal stock movements efficiently.
- Developed dynamic inbound and outbound warehouse processes with a hybrid FEFO/FIFO stock allocation algorithm, improving inventory rotation and shelf-life management.
- Engineered an automated inventory pipeline using Laravel Jobs and cron, running 24/7 to ensure 100% detection of low-stock and near-expiration batches with proactive email alerts.

SecureSign (Digital Signature System) | Live Demo | GitHub

Apr 2026 – Present

Tech: FastAPI, PostgreSQL, Supabase, SQLAlchemy, Tailwind

- Built a full-stack web tool for uploading PDF files, generating embedded digital signatures (PAdES standard), and validating document authenticity.
- Applied RSA cryptography algorithm for secure key generation to prevent document tampering.
- Utilized FastAPI and SQLAlchemy to handle user signing flows, integrated with Supabase for robust cloud storage and deployment compatibility.
- Implemented event logging via PostgreSQL JSONB to track user actions and cryptographic audit trails.
- Implemented CI/CD pipelines and ensured backend reliability with 81% test coverage using PyTest.

POS Management System | Live Demo | GitHub

Mar 2025 – May 2025

Tech: React.js, Vite, NestJS, TypeORM, MySQL, JWT, Tailwind

- Developed a full-stack Point of Sale (POS) system for restaurant operations, including order management, table reservations, and QR-based ordering.
- Implemented RESTful APIs, automated invoice generation, and state-driven table lifecycle management using NestJS, TypeORM, and MySQL.

SmartDoc AI (Local Document RAG) | GitHub

Mar 2026 – May 2026

Tech: Python, LangChain, FAISS, Ollama

- Developed a local Retrieval-Augmented Generation (RAG) system for offline document search and question answering using LangChain, FAISS, and Ollama.

Open Source Contributions

Railpack (Railway) | Open Source Contributor | Issue #610

Jun 2026

- Authored a core-engine PR in Go (successfully merged) to resolve a configuration bug, ensuring the runtime version resolution strictly follows the documented precedence rules.
- Collaborated directly with maintainers across multiple production releases (v0.29.0, v0.30.0) by contributing reproducible examples (MREs), root-cause analyses, and architectural reviews.